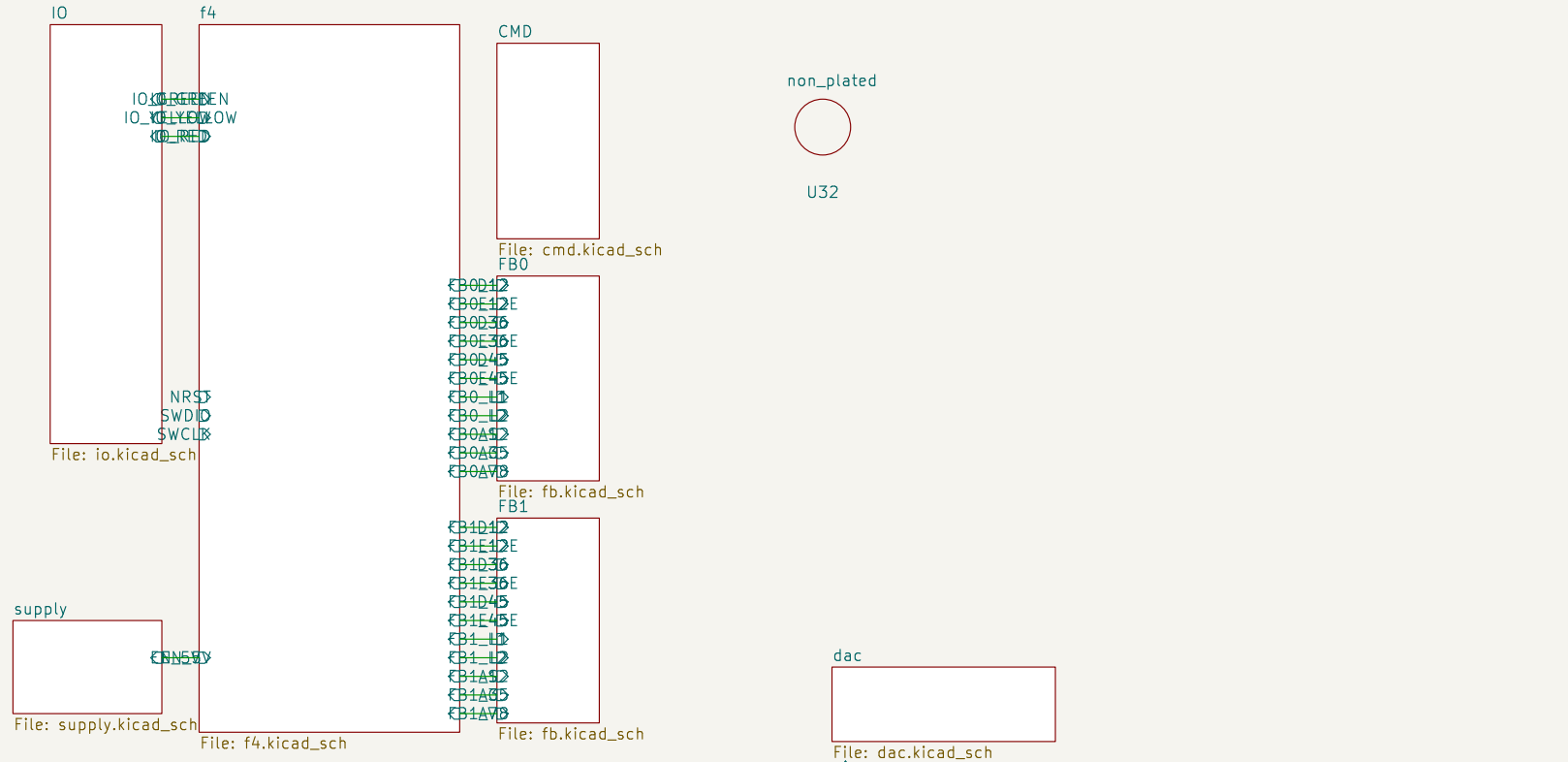
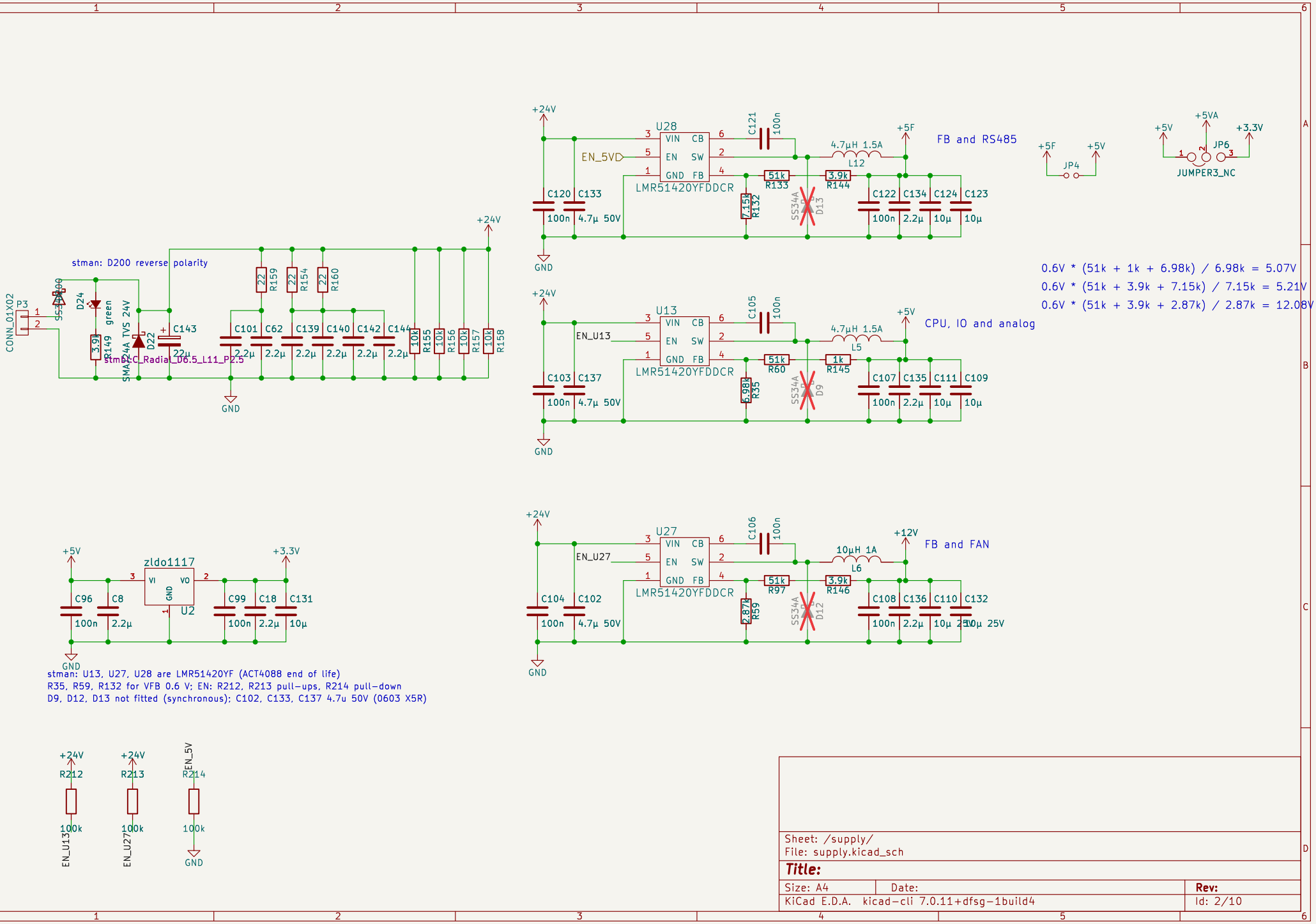


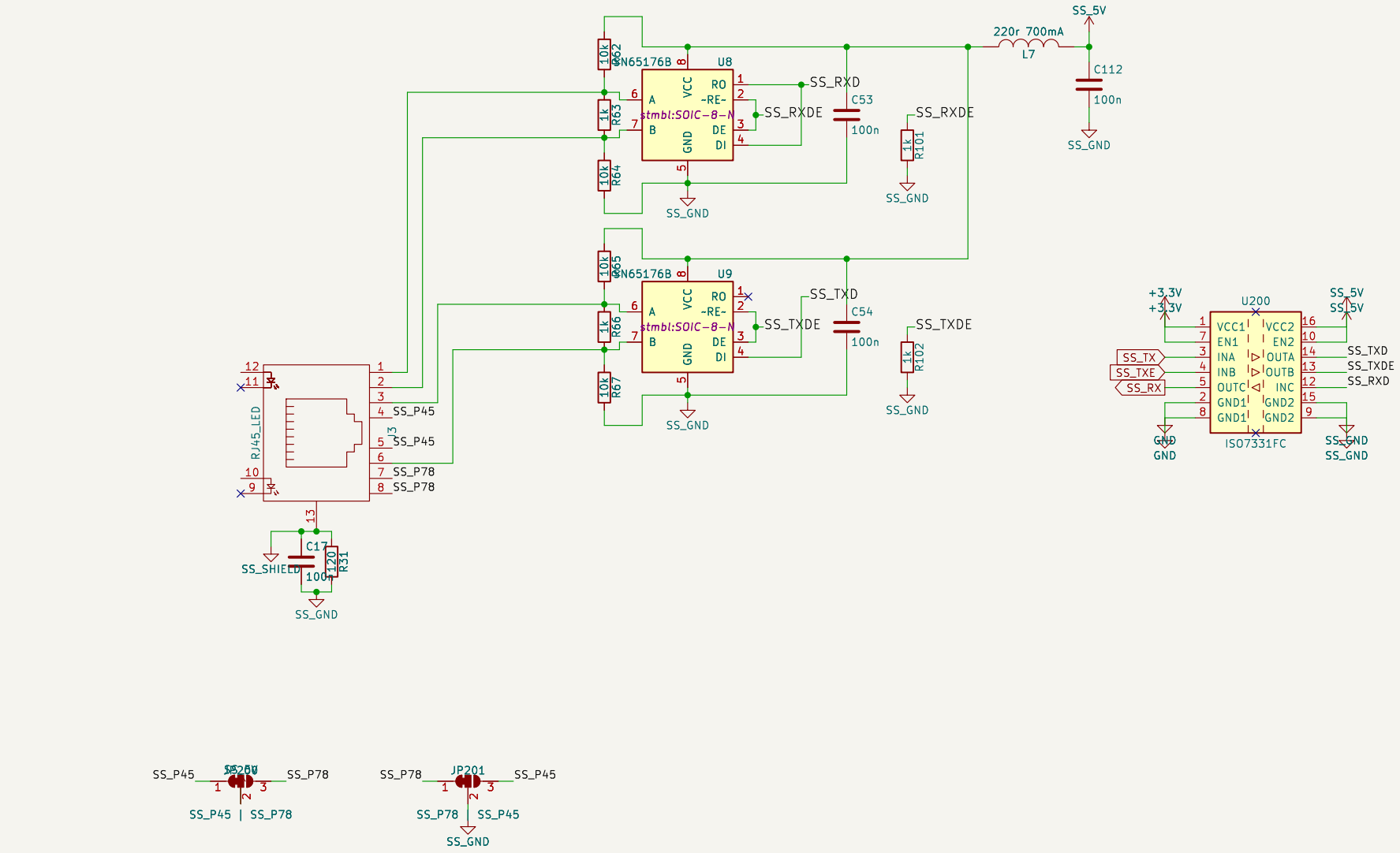
stman (docs/stman.md): stmbl v4.0's LV sheets, edited in place;
dac, ain, drvio: stman's new circuits (global labels to the MCU sheet).

Sheet: /
File: stman.kicad_sch

Title: stman: stmbl with an analog output

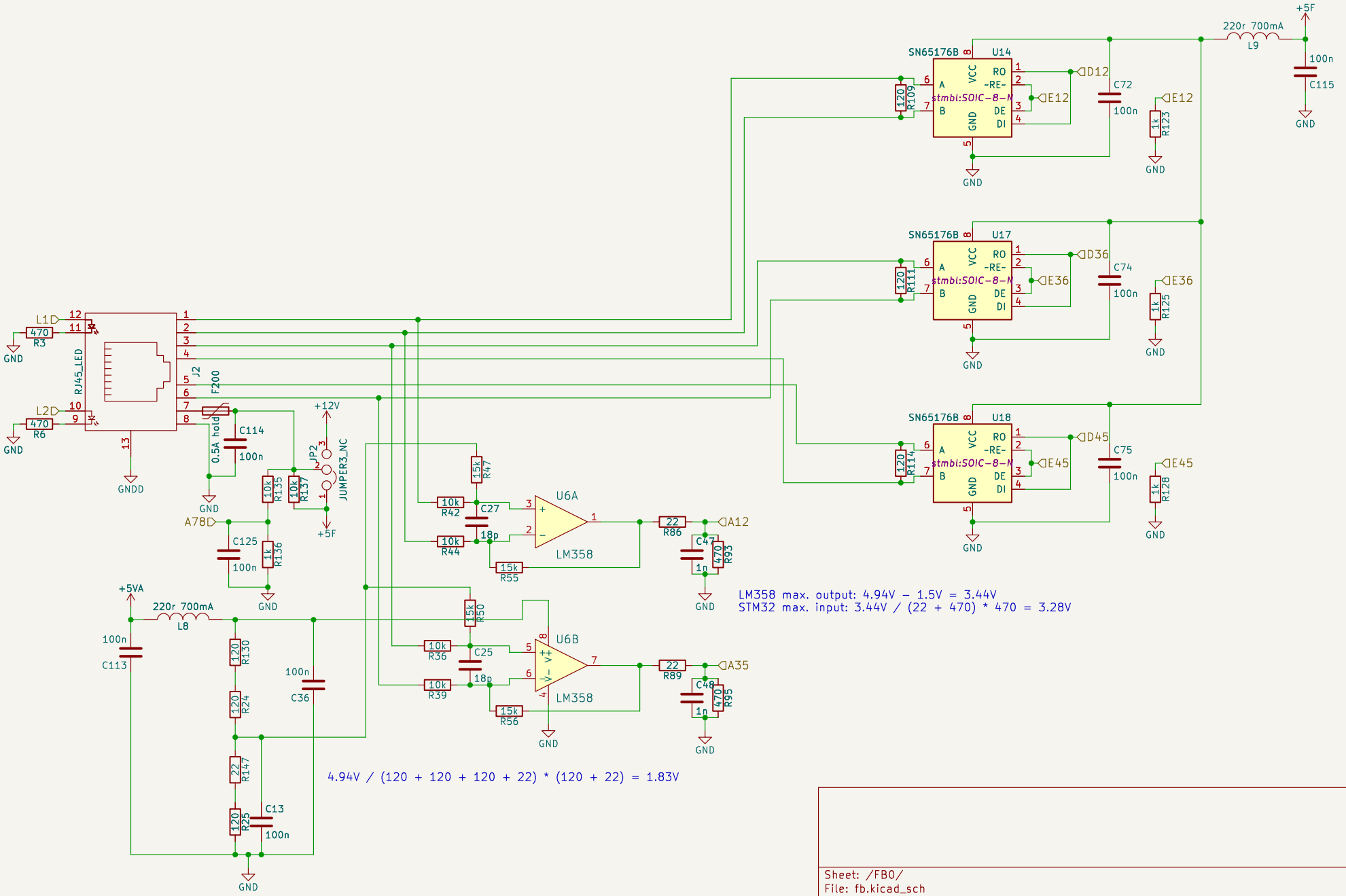
Size: A4	Date:	Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/10

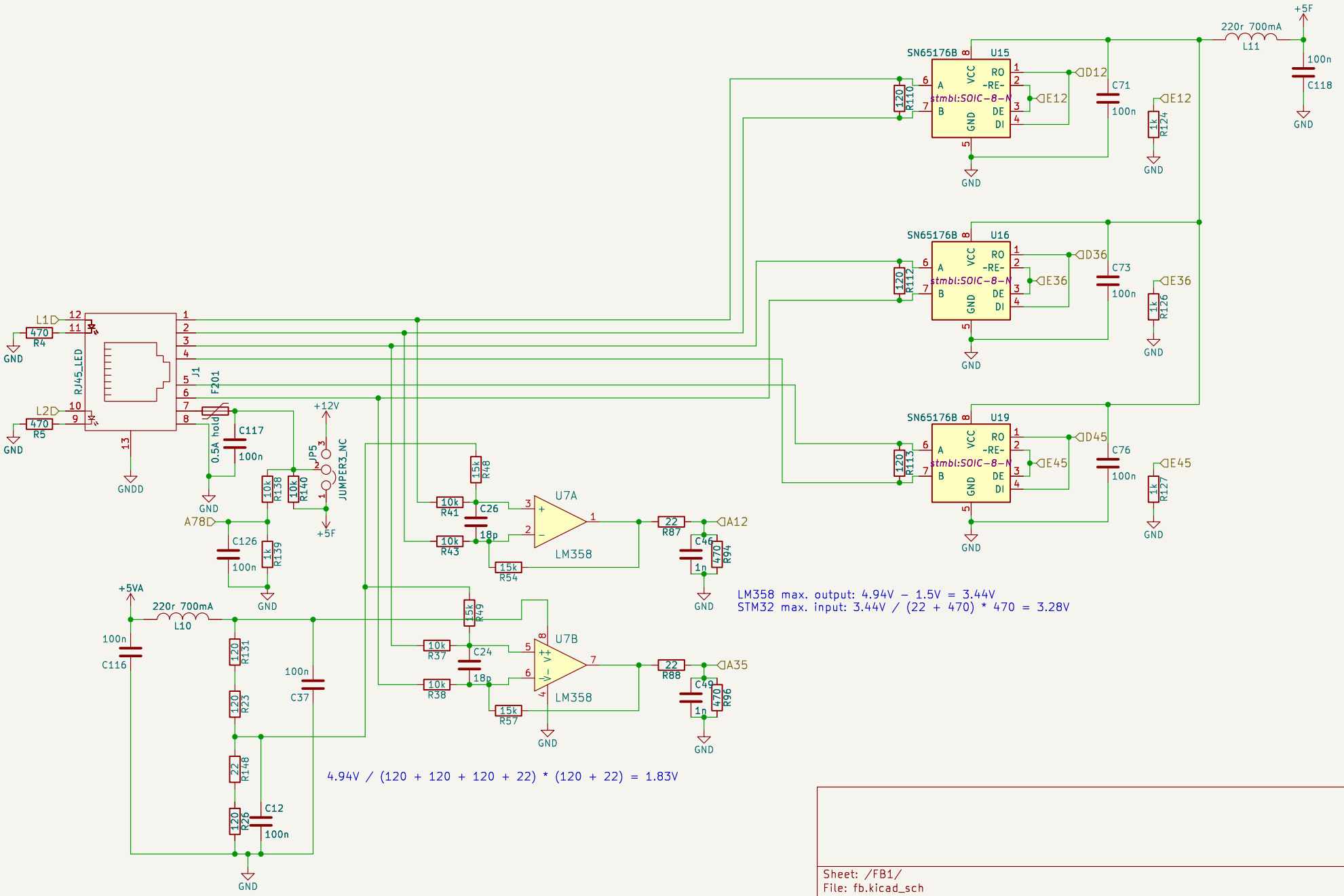

stman: D200 reverse polarity
CONN_01X02
P3
SS34A
D24
green
3.9
R149
SMAJ24A TVS 24V
D22
C143
22µ
stman:CP_Radial_D6.5_L11_P2.5
22
R159
22
R154
22
R160
C101 C62 C139 C140 C142 C144
2.2µ
10k
R155
10k
R156
10k
R157
10k
R158
+24V
GND
U28
LMR51420YFDDCR
VIN CB
EN SW
GND FB
EN_5VD
C121
100n
4.7µH 1.5A
L12
+5F
FB and RS485
51k
R133
3.9k
R144
7.15k
R132
C120 C133
100n 4.7µ 50V
C122 C134 C124 C123
100n 2.2µ 10µ 10µ
U13
EN_U13
C105
L5
CPU, IO and analog
+5V
1k
R145
R60
6.98k
R35
C103 C137
C107 C135 C111 C109
U27
EN_U27
C106
10µH 1A
L6
+12V
FB and FAN
R97
R146
2.87k
R59
C104 C102
C108 C136 C110 C132
25V
zldo1117
VI VO GND
U2
+5V
+3.3V
C96 C8
C99 C18 C131
100n 2.2µ 10µ
JP4
JP6
JUMPER3_NC
+5VA
+3.3V
0.6V * (51k + 1k + 6.98k) / 6.98k = 5.07V
0.6V * (51k + 3.9k + 7.15k) / 7.15k = 5.21V (stman
0.6V * (51k + 3.9k + 2.87k) / 2.87k = 12.08V
stman: U13, U27, U28 are LMR51420YF (ACT4088 end of life)
R35, R59, R132 for VFB 0.6 V; EN: R212, R213 pull-ups, R214 pull-down
D9, D12, D13 not fitted (synchronous); C102, C133, C137 4.7u 50V (0603 X5R)
R212
R213
R214
EN_5V
100k
EN_U13
EN_U27
Sheet: /supply/
File: supply.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 2/10

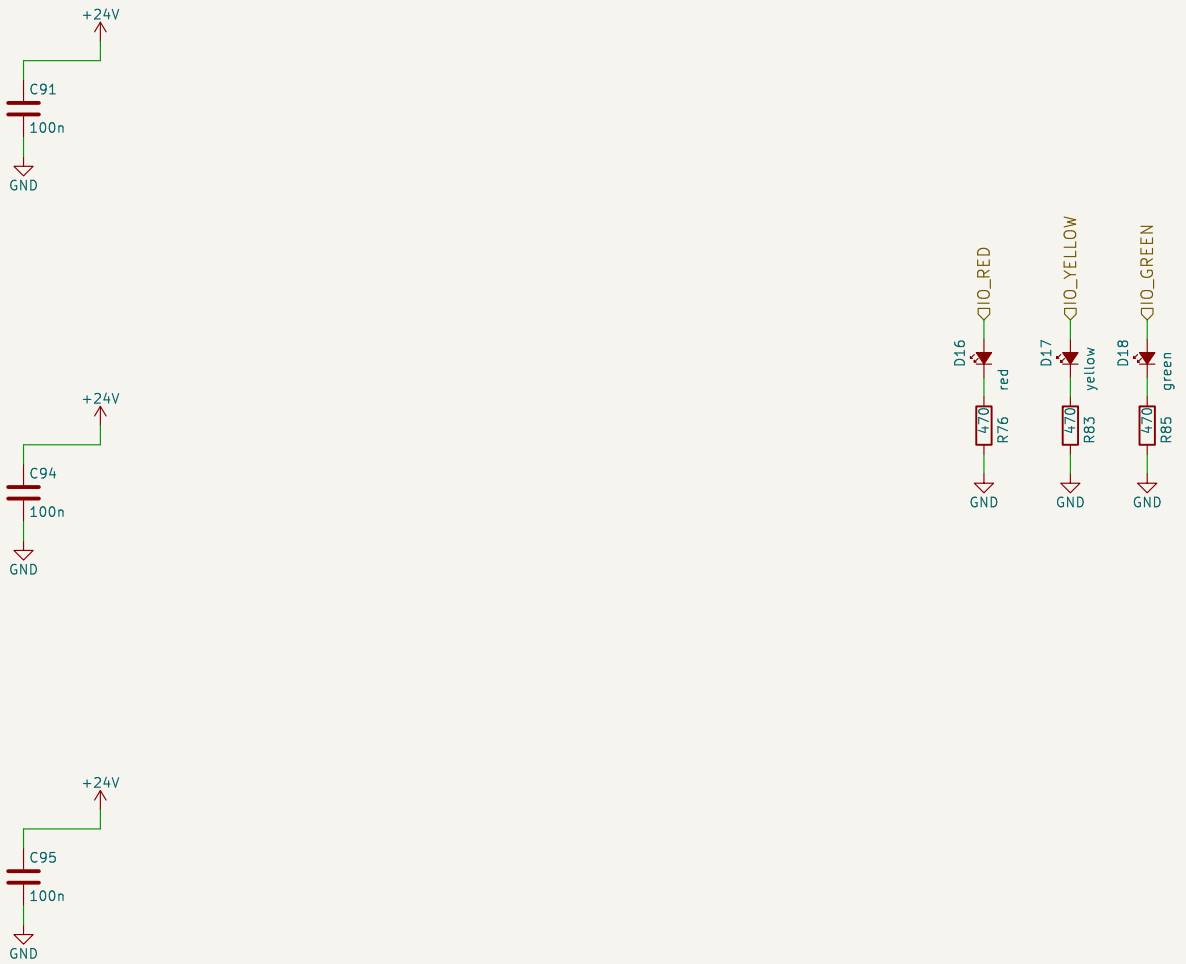


stman: the transceivers U8 (receive, pair 1/2) and U9 (send, pair 3/6) are on the cable's side,
supplied from the cable (SS_5V, SS_GND); ISO7331FC U200 (outputs low while unpowered) is the only path to the MCU.
U10, U11 (pairs 4/5, 7/8) removed: those pairs carry the cable's +5 V and GND (JP200, JP201: check against the Mesa card *)

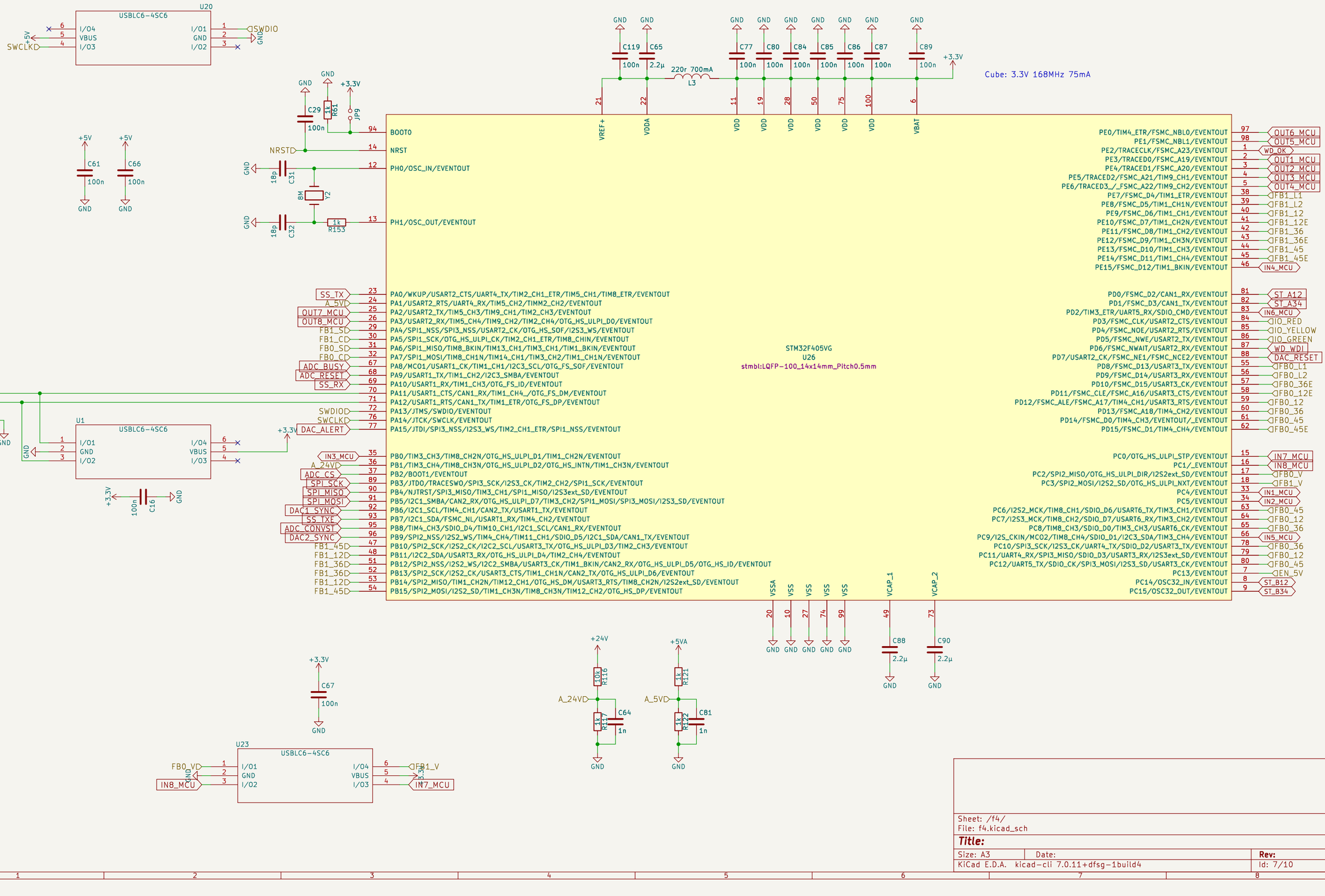
Sheet: /CMD/ File: cmd.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/10


SN65176B
U14
U17
U18
stmbl:SOIC-8-N
VCC
GND
A
B
RO
-RE-
DE
DI
120
R109
R111
R114
1k
R123
R125
R128
D12
D36
D45
E12
E36
E45
C72
C74
C75
100n
220r 700mA
L9
C115
+5F
GND
L1
L2
470
R3
R6
RJ45_LED
J2
F200
0.5A hold
C114
GNDD
+12V
JP2
JUMPER3_NC
10k
R135
R137
A78
C125
1k
R136
15k
R47
R42
C27
18p
R44
R55
U6A
LM358
22
R86
A12
C47
470
R93
1n
R50
R36
R39
R56
U6B
R89
A35
C48
R95
+5VA
L8
C113
R130
R24
R147
R25
C13
C36
LM358 max. output: 4.94V − 1.5V = 3.44V
STM32 max. input: 3.44V / (22 + 470) * 470 = 3.28V
4.94V / (120 + 120 + 120 + 22) * (120 + 22) = 1.83V
Sheet: /FB0/
File: fb.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 4/10

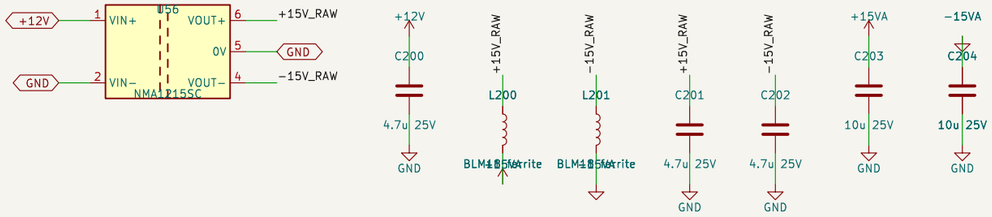

SN65176B
U15
U16
U19
stmbl:SOIC-8-N
VCC
GND
A
B
RO
-RE-
DE
DI
D12
D36
D45
E12
E36
E45
C71
C73
C76
100n
220r 700mA
L11
C118
+5F
GND
120
R110
R112
R113
1k
R124
R126
R127
J1
RJ45_LED
F201
0.5A hold
L1
L2
470
R4
R5
C117
GNDD
+12V
JP5
JUMPER3_NC
A78
10k
R138
R140
C126
R139
U7A
U7B
LM358
15k
R48
R49
R54
R41
R43
R37
R38
C26
C24
18p
22
R87
R88
C46
C49
1n
R94
R96
A12
A35
+5VA
L10
C116
C37
R131
R23
R148
R26
C12
LM358 max. output: 4.94V − 1.5V = 3.44V
STM32 max. input: 3.44V / (22 + 470) * 470 = 3.28V
4.94V / (120 + 120 + 120 + 22) * (120 + 22) = 1.83V
Sheet: /FB1/
File: fb.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 5/10



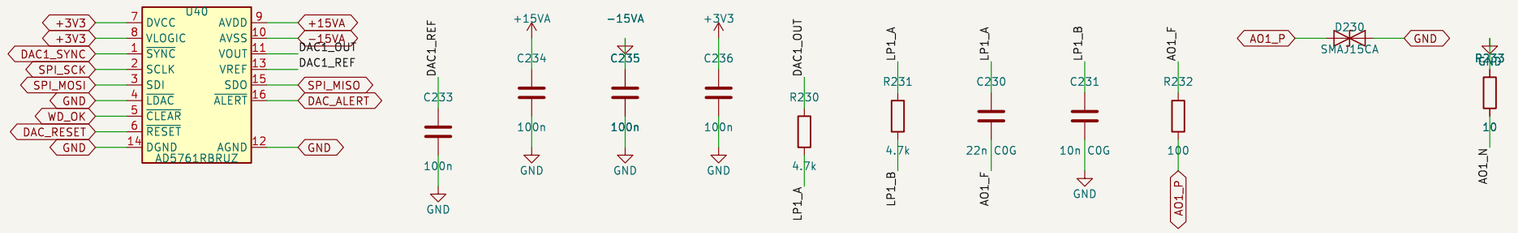
Sheet: /IO/		
File: io.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/10


U20
USBLC6-4SC6
I/O4
VBUS
I/O3
I/O1
GND
I/O2
SWCLK
SWDIO
+5V
GND
C61
C66
100n
C29
R61
1k
JP9
NRST
C31
C32
18p
8M
Y2
R153
BOOT0
NRST
PH0/OSC_IN/EVENTOUT
PH1/OSC_OUT/EVENTOUT
C119
C65
2.2µ
220r 700mA
L3
C77
C80
C84
C85
C86
C87
C89
+3.3V
Cube: 3.3V 168MHz 75mA
VREF+
VDDA
VDD
VBAT
STM32F405VG
U26
stmbl:LQFP-100_14x14mm_Pitch0.5mm
PA0/WKUP/USART2_CTS/UART4_TX/TIM2_CH1_ETR/TIM5_CH1/TIM8_ETR/EVENTOUT
PA1/USART2_RTS/UART4_RX/TIM5_CH2/TIMM2_CH2/EVENTOUT
PA2/USART2_TX/TIM5_CH3/TIM9_CH1/TIM2_CH3/EVENTOUT
PA3/USART2_RX/TIM5_CH4/TIM9_CH2/TIM2_CH4/OTG_HS_ULPI_D0/EVENTOUT
PA4/SPI1_NSS/SPI3_NSS/USART2_CK/OTG_HS_SOF/I2S3_WS/EVENTOUT
PA5/SPI1_SCK/OTG_HS_ULPI_CK/TIM2_CH1_ETR/TIM8_CHIN/EVENTOUT
PA6/SPI1_MISO/TIM8_BKIN/TIM13_CH1/TIM3_CH1/TIM1_BKIN/EVENTOUT
PA7/SPI1_MOSI/TIM8_CH1N/TIM14_CH1/TIM3_CH2/TIM1_CH1N/EVENTOUT
PA8/MCO1/USART1_CK/TIM1_CH1/I2C3_SCL/OTG_FS_SOF/EVENTOUT
PA9/USART1_TX/TIM1_CH2/I2C3_SMBA/EVENTOUT
PA10/USART1_RX/TIM1_CH3/OTG_FS_ID/EVENTOUT
PA11/USART1_CTS/CAN1_RX/TIM1_CH4_/OTG_FS_DM/EVENTOUT
PA12/USART1_RTS/CAN1_TX/TIM1_ETR/OTG_FS_DP/EVENTOUT
PA13/JTMS/SWDIO/EVENTOUT
PA14/JTCK/SWCLK/EVENTOUT
PA15/JTDI/SPI3_NSS/I2S3_WS/TIM2_CH1_ETR/SPI1_NSS/EVENTOUT
PB0/TIM3_CH3/TIM8_CH2N/OTG_HS_ULPI_D1/TIM1_CH2N/EVENTOUT
PB1/TIM3_CH4/TIM8_CH3N/OTG_HS_ULPI_D2/OTG_HS_INTN/TIM1_CH3N/EVENTOUT
PB2/BOOT1/EVENTOUT
PB3/JTDO/TRACESWO/SPI3_SCK/I2S3_CK/TIM2_CH2/SPI1_SCK/EVENTOUT
PB4/NJTRST/SPI3_MISO/TIM3_CH1/SPI1_MISO/I2S3ext_SD/EVENTOUT
PB5/I2C1_SMBA/CAN2_RX/OTG_HS_ULPI_D7/TIM3_CH2/SPI1_MOSI/SPI3_MOSI/I2S3_SD/EVENTOUT
PB6/I2C1_SCL/TIM4_CH1/CAN2_TX/USART1_TX/EVENTOUT
PB7/I2C1_SDA/FSMC_NL/USART1_RX/TIM4_CH2/EVENTOUT
PB8/TIM4_CH3/SDIO_D4/TIM10_CH1/I2C1_SCL/CAN1_RX/EVENTOUT
PB9/SPI2_NSS/I2S2_WS/TIM4_CH4/TIM11_CH1/SDIO_D5/I2C1_SDA/CAN1_TX/EVENTOUT
PB10/SPI2_SCK/I2S2_CK/I2C2_SCL/USART3_TX/OTG_HS_ULPI_D3/TIM2_CH3/EVENTOUT
PB11/I2C2_SDA/USART3_RX/OTG_HS_ULPI_D4/TIM2_CH4/EVENTOUT
PB12/SPI2_NSS/I2S2_WS/I2C2_SMBA/USART3_CK/TIM1_BKIN/CAN2_RX/OTG_HS_ULPI_D5/OTG_HS_ID/EVENTOUT
PB13/SPI2_SCK/I2S2_CK/USART3_CTS/TIM1_CH1N/CAN2_TX/OTG_HS_ULPI_D6/EVENTOUT
PB14/SPI2_MISO/TIM1_CH2N/TIM12_CH1/OTG_HS_DM/USART3_RTS/TIM8_CH2N/I2S2ext_SD/EVENTOUT
PB15/SPI2_MOSI/I2S2_SD/TIM1_CH3N/TIM8_CH3N/TIM12_CH2/OTG_HS_DP/EVENTOUT
PC0/OTG_HS_ULPI_STP/EVENTOUT
PC1/_EVENTOUT
PC2/SPI2_MISO/OTG_HS_ULPI_DIR/I2S2ext_SD/EVENTOUT
PC3/SPI2_MOSI/I2S2_SD/OTG_HS_ULPI_NXT/EVENTOUT
PC4/EVENTOUT
PC5/EVENTOUT
PC6/I2S2_MCK/TIM8_CH1/SDIO_D6/USART6_TX/TIM3_CH1/EVENTOUT
PC7/I2S3_MCK/TIM8_CH2/SDIO_D7/USART6_RX/TIM3_CH2/EVENTOUT
PC8/TIM8_CH3/SDIO_D0/TIM3_CH3/USART6_CK/EVENTOUT
PC9/I2S_CKIN/MCO2/TIM8_CH4/SDIO_D1/I2C3_SDA/TIM3_CH4/EVENTOUT
PC10/SPI3_SCK/I2S3_CK/UART4_TX/SDIO_D2/USART3_TX/EVENTOUT
PC11/UART4_RX/SPI3_MISO/SDIO_D3/USART3_RX/I2S3ext_SD/EVENTOUT
PC12/UART5_TX/SDIO_CK/SPI3_MOSI/I2S3_SD/USART3_CK/EVENTOUT
PC13/EVENTOUT
PC14/OSC32_IN/EVENTOUT
PC15/OSC32_OUT/EVENTOUT
PD0/FSMC_D2/CAN1_RX/EVENTOUT
PD1/FSMC_D3/CAN1_TX/EVENTOUT
PD2/TIM3_ETR/UART5_RX/SDIO_CMD/EVENTOUT
PD3/FSMC_CLK/USART2_CTS/EVENTOUT
PD4/FSMC_NOE/USART2_RTS/EVENTOUT
PD5/FSMC_NWE/USART2_TX/EVENTOUT
PD6/FSMC_NWAIT/USART2_RX/EVENTOUT
PD7/USART2_CK/FSMC_NE1/FSMC_NCE2/EVENTOUT
PD8/FSMC_D13/USART3_TX/EVENTOUT
PD9/FSMC_D14/USART3_RX/EVENTOUT
PD10/FSMC_D15/USART3_CK/EVENTOUT
PD11/FSMC_CLE/FSMC_A16/USART3_CTS/EVENTOUT
PD12/FSMC_ALE/FSMC_A17/TIM4_CH1/USART3_RTS/EVENTOUT
PD13/FSMC_A18/TIM4_CH2/EVENTOUT
PD14/FSMC_D0/TIM4_CH3/EVENTOUT/_EVENTOUT
PD15/FSMC_D1/TIM4_CH4/EVENTOUT
PE0/TIM4_ETR/FSMC_NBL0/EVENTOUT
PE1/FSMC_NBL1/EVENTOUT
PE2/TRACECLK/FSMC_A23/EVENTOUT
PE3/TRACED0/FSMC_A19/EVENTOUT
PE4/TRACED1/FSMC_A20/EVENTOUT
PE5/TRACED2/FSMC_A21/TIM9_CH1/EVENTOUT
PE6/TRACED3_/_FSMC_A22/TIM9_CH2/EVENTOUT
PE7/FSMC_D4/TIM1_ETR/EVENTOUT
PE8/FSMC_D5/TIM1_CH1N/EVENTOUT
PE9/FSMC_D6/TIM1_CH1/EVENTOUT
PE10/FSMC_D7/TIM1_CH2N/EVENTOUT
PE11/FSMC_D8/TIM1_CH2/EVENTOUT
PE12/FSMC_D9/TIM1_CH3N/EVENTOUT
PE13/FSMC_D10/TIM1_CH3/EVENTOUT
PE14/FSMC_D11/TIM1_CH4/EVENTOUT
PE15/FSMC_D12/TIM1_BKIN/EVENTOUT
VSSA
VSS
VCAP_1
VCAP_2
C88
C90
2.2µ
SS_TX
A_5V
OUT7_MCU
OUT8_MCU
FB1_SD
FB1_CD
FB0_SD
FB0_CD
ADC_BUSY
ADC_RESET
SS_RX
SWDIO
SWCLK
DAC_ALERT
IN3_MCU
A_24V
ADC_CS
SPI_SCK
SPI_MISO
SPI_MOSI
DAC1_SYNC
SS_TXE
ADC_CONVST
DAC2_SYNC
FB1_45
FB1_12
FB1_36
FB1_36
FB1_12
FB1_45
OUT6_MCU
OUT5_MCU
WD_OK
OUT1_MCU
OUT2_MCU
OUT3_MCU
OUT4_MCU
FB1_L1
FB1_L2
FB1_12
FB1_12E
FB1_36
FB1_36E
FB1_45
FB1_45E
IN4_MCU
ST_A12
ST_A34
IN6_MCU
IO_RED
IO_YELLOW
IO_GREEN
WD_WDI
DAC_RESET
FB0_L1
FB0_L2
FB0_36E
FB0_12E
FB0_12
FB0_36
FB0_45
FB0_45E
IN7_MCU
IN8_MCU
FB0_V
FB1_V
IN1_MCU
IN2_MCU
FB0_45
FB0_12
FB0_36
IN5_MCU
FB0_36
FB0_12
FB0_45
EN_5V
ST_B12
ST_B34
USB_OTG
VCC
D-
D+
ID
GND
shield
P1
C1
100n
120
R1
U1
USBLC6-4SC6
C16
U23
FB0_V
IN8_MCU
FB1_V
IN7_MCU
C67
+24V
+5VA
A_24V
A_5V
10Ω
R116
1k
R117
C64
1n
R121
R122
C81
Sheet: /f4/
File: f4.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 7/10

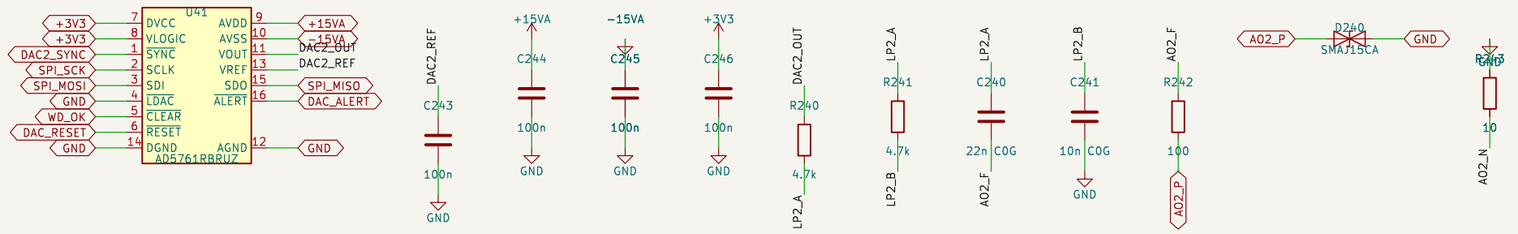
±15 V: NMA1215 from +12 V, LC filtered



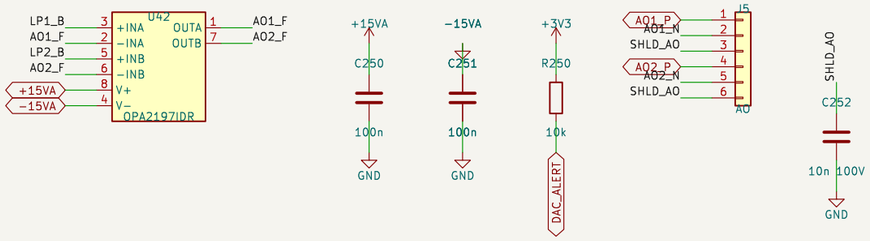
AO1: DAC, Sallen–Key (R230 R231 C230 C231, U42A), 100 R, TVS



AO2 (fitted for the spindle's C axis): DAC, Sallen–Key (U42B)



Filter op–amp, /ALERT, connector J5



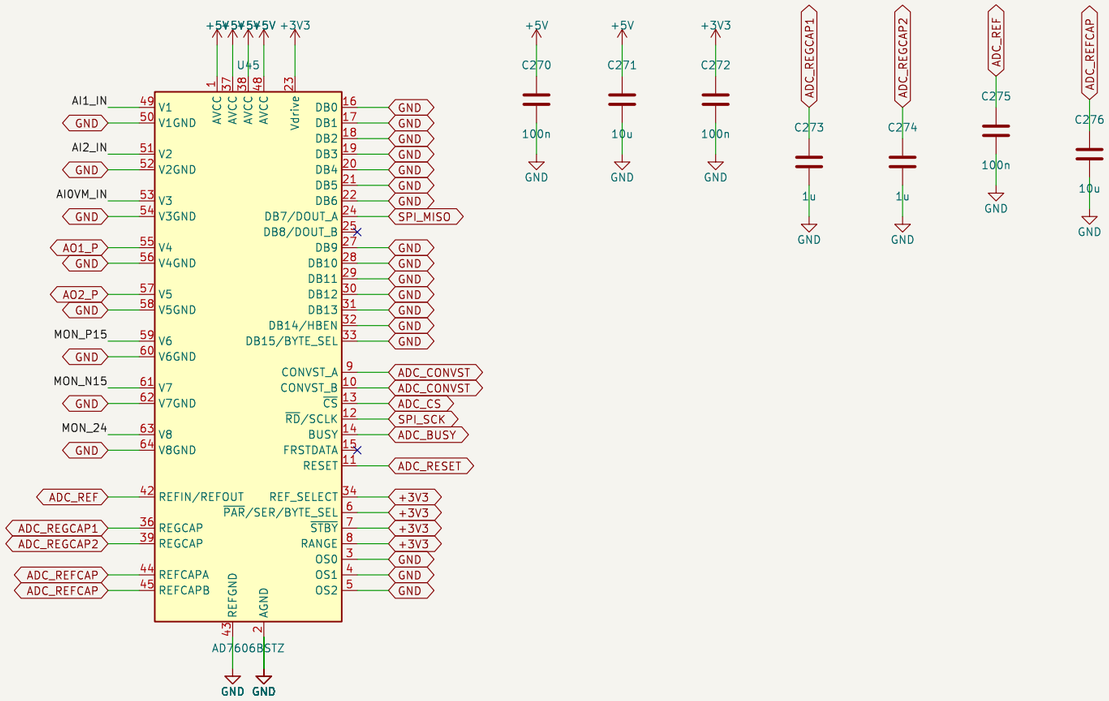
* in a part's Note: check against the datasheet
docs/stman.md

Sheet: /dac/
File: dac.kicad_sch

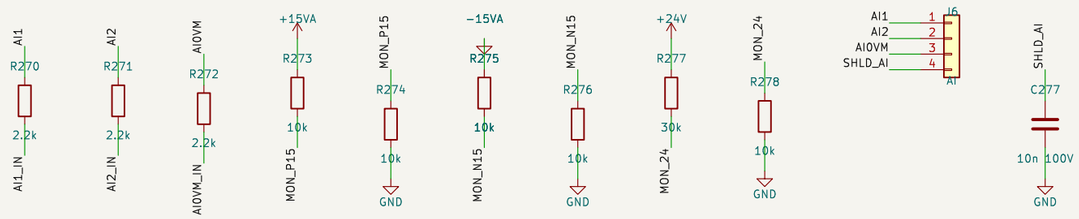
Title: stman: Command outputs AO1, AO2: AD5761R, 2nd–order low–pass 2.3

Size: A3	Date:	**Rev:**
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AD7606: 8 ch simultaneous, ±10 V (RANGE high), serial (PAR/SER high), internal reference



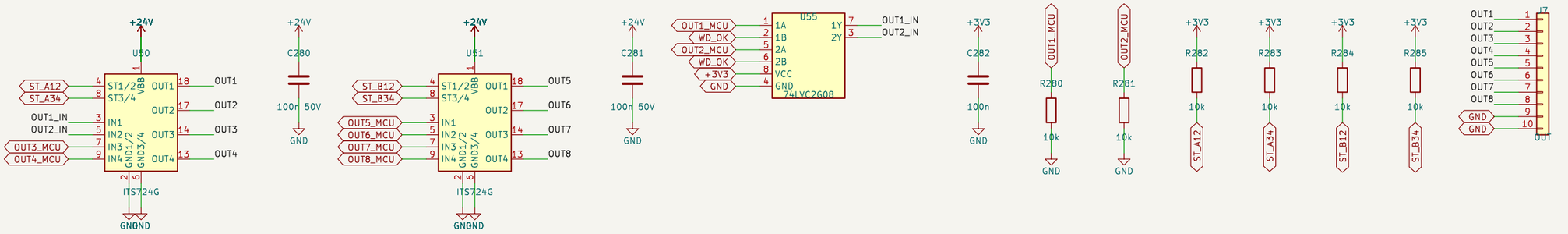
Inputs: 2.2k series (the ADC's 1 MΩ inputs, ±16.5 V clamps); rail dividers; J6



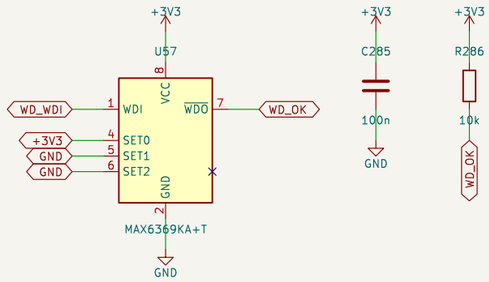
* in a part's Note: check against the datasheet
docs/stman.md

Sheet: /ain/		
File: ain.kicad_sch		
Title: stman: Monitor ADC: AD7606 (drive monitors and 0VM, AO readbacks, ra		
Size: A3	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 11/10

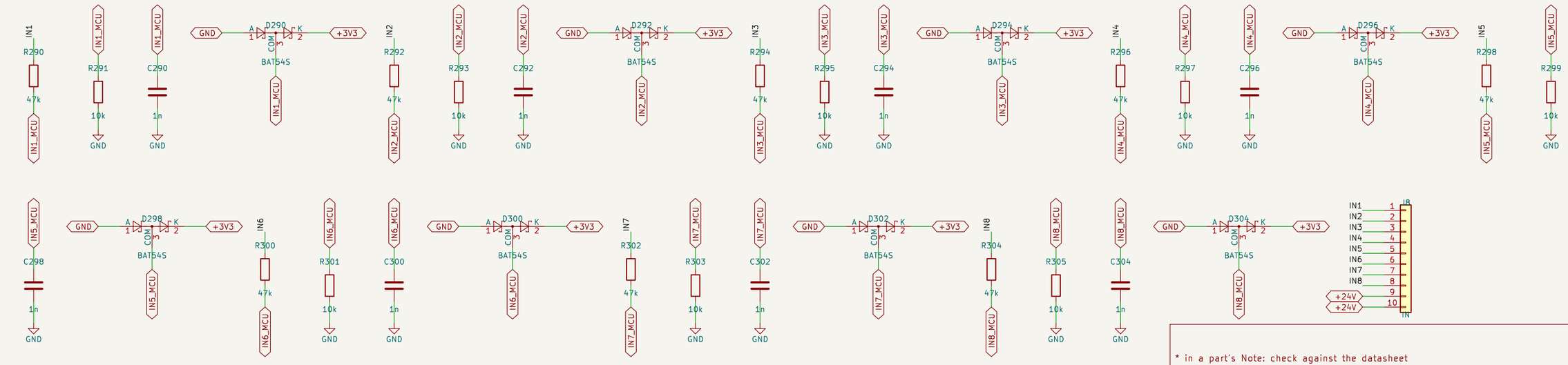
Outputs: ITS724G x2, OUT1/OUT2 gated by the watchdog



Watchdog MAX6369: WD_OK low -> DACs to 0 V, OUT1/OUT2 off



Inputs: 47k / 10k, 1 nF, BAT54S clamp (high > ~13 V, low < ~6 V)



* in a part's Note: check against the datasheet
docs/stman.md

Sheet: /drvio/
File: drvio.kicad_sch

Title: stman: Drive I/O: 8 x 24 V out (ITS724G), 8 x 24 V in, watchdog

Size: A3	Date:	Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 12/10